

ETL Pipeline Execution Report

1. Project Summary

- **Project Name:** Banking_Data_Pipeline
 - **Main Script:** src/main.py
 - **Date:** August 11, 2026
 - **Total Run Time:** 7.28 seconds
 - **Status:** SUCCESS
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2. Where the Data Went (Data Flow)

The pipeline processed every transaction step-by-step without stopping:

- **Started with:** 300,000 raw rows from 3 regional files.
 - **Bad Data Found:** 2,063 rows were sent to quarantine.
 - **Good Data Remaining:** 297,937 rows were cleaned.
 - **Suspicious Data Found:** 148,658 rows were marked as risky fraud.
 - **Saved to Clean File:** 149,279 rows were completely safe and saved.
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3. Step-by-Step Details

Pipeline Execution (src/main.py)

- **Action:** This script serves as the central controller for the entire ETL pipeline.
- **Execution:** Running the command `python src/main.py` in the terminal automatically triggers and coordinates all the subsequent modules (Step 1 to Step 5) in the correct sequence.

Step 1: Extract (src/extract.py)

- The script checked and loaded 3 regional files successfully:
 - **North Records:** 100,000
 - **South Records:** 100,000
 - **West Records:** 100,000

- **Total Collected: 300,000 records**

Step 2: Validate & Quarantine (src/validate.py)

- The script checked all 300,000 records for mistakes. It found **2,063 bad/invalid records** and saved them separately inside *quarantine/quarantine_records.csv*.
- **The mistakes found were:**
 - Missing Account Numbers: **607 rows**
 - Missing Customer IDs: **592 rows**
 - Negative Money Amounts: **865 rows**
 - Duplicates or Future Dates: **0**

Step 3: Transform (src/transform.py)

- The script cleaned the **297,937 good records**:
 - Fixed dates into standard datetime formats.
 - Turned account numbers into text/strings.
 - Checked for identical rows (**0 duplicate rows were found**).
 - Sorted everything in order by date.

Step 4: Fraud Detection (src/fraud.py)

- The script looked at the clean records to find dangerous activity (like transactions over \$10,000).
- It flagged **148,658 rows as suspicious** and saved them separately inside *fraud_data/fraud_transactions.csv*.

Step 5: Load (src/load.py)

- The final **149,279 perfectly safe records** were saved directly into *processed/clean_transactions.csv*.

Pipeline Execution Output:

The following screenshot shows the complete terminal logs after running **main.py**, displaying the successful execution of the ETL pipeline stages.

```
File Edit Selection View ... Banking_Data_Pipeline
src > main.py
1 import time
2 from logger import logger # Import our central ETL logger
3 from extract import extract_records, north_file_path, south_file_path, west_file_path
...
PS D:\Banking_Data_Pipeline> python src/main.py
2026-08-11 09:42:40 - INFO - ETL Started

-----
Extraction Started
Total Files Loaded : 3
North Records : 100000
South Records : 100000
West Records : 100000
Total Records Extracted : 300000
Total records extracted successfully: 300000
--- Extraction completed ---
2026-08-11 09:42:41 - INFO - Extraction Completed - Processed: 300000 rows

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Validation Started
Total Missing Account Number Count: 607
Total Missing Customer ID count: 592
Total Negative Amount: 865
No duplicated transaction id found
No future transaction date
Transaction types are valid
All channel types are valid
No Missing or null branch code

Validation Summary
Total Input Records : 300000
Total Valid Records : 297937
Total Invalid Records : 2063
Validation Completed
2026-08-11 09:42:42 - INFO - Validation Completed - Valid: 297937, Invalid: 2063

-----
--- Quarantine Processing Started ---
Total Invalid Records 2063
Generating reasons for quarantine records...
Quarantine Dataset Saved Successfully at: D:\Banking_Data_Pipeline\quarantine\quarantine_records.csv
Total Quarantine Records Saved: 2063
--- Quarantine Processing Ended ---
2026-08-11 09:42:42 - INFO - Quarantine Completed - 2063

-----
```

Figure 3.1: Terminal output for main.py execution (Step 1 to Step 2)

```
2026-08-11 09:42:42 - INFO - Validation Completed - Valid: 297937, Invalid: 2063

-----
--- Quarantine Processing Started ---
Total Invalid Records 2063
Generating reasons for quarantine records...
Quarantine Dataset Saved Successfully at: D:\Banking_Data_Pipeline\quarantine\quarantine_records.csv
Total Quarantine Records Saved: 2063
--- Quarantine Processing Ended ---
2026-08-11 09:42:42 - INFO - Quarantine Completed - 2063

-----
--- Transformation of Data Started ---
Initial Records : 297937
Convert transaction date to proper datetime format
Convert account number to string format
Removing duplicate records
Final records after removing 0 duplicated rows: 297937
Sorting records by transaction date...
Final records after transformation: 297937
--- Transformation of Data Completed ---
2026-08-11 09:42:45 - INFO - Transformation Completed

-----
--- Fraud Detection Started ---
Total Scanned Rows : 297937
Total Fraud Transactions Found : 148658
Total Clean Records Found : 149279
Fraud Transactions Saved At : D:\Banking_Data_Pipeline\fraud_data\fraud_transactions.csv
--- Fraud Detection Ended ---
2026-08-11 09:42:46 - INFO - Fraud Detection Completed - Flagged Anomalies: 148658

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--- Clean Data Loading Started ---
Successfully Loaded 149279 Clean Records to csv
--- Data Loading Ended ---
2026-08-11 09:42:47 - INFO - Data Loaded Successfully

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2026-08-11 09:42:47 - INFO - ETL Finished successfully in 7.2816 seconds

PS D:\Banking_Data_Pipeline>
```

Figure 3.2: Terminal output for main.py execution (Step 2 to Step 5)

4. Pipeline Timeline Logs (logs/etl.log)

This timeline shows exactly what time each step finished in your terminal:

2026-08-11 09:42:40 - INFO - ETL Started

2026-08-11 09:42:41 - INFO - Extraction Completed - Processed: 300000 rows

2026-08-11 09:42:42 - INFO - Validation Completed - Valid: 297937, Invalid: 2063

2026-08-11 09:42:42 - INFO - Quarantine Completed - 2063

2026-08-11 09:42:45 - INFO - Transformation Completed

2026-08-11 09:42:46 - INFO - Fraud Detection Completed - Flagged Anomalies: 148658

2026-08-11 09:42:47 - INFO - Data Loaded Successfully

2026-08-11 09:42:47 - INFO - ETL Finished successfully in 7.2816 seconds

5. Key Points to Know

i. Error Rate Formula & Calculation

Total Input Rows (Total): 300,000

Bad/Invalid Rows (Part): 2,063

- Formula: $(\text{Total Invalid Records} / \text{Total Input Records}) * 100$
- Math calculation: $(2,063 / 300,000) * 100$
- Result: 0.6876% (rounds up to 0.69%)

ii. Fraud Flags Formula & Calculation

Total Valid Records: 297,937

Total Fraud Records: 148,658

- Formula: $(\text{Total Fraud Records} / \text{Total Valid Records}) * 100$
- Math calculation: $(148,658 / 297,937) * 100$
- Result: 49.8957% (rounds up to 49.89%)

iii. **Fast Performance:** The pipeline processed 300,000 records and saved 3 files in **under 8 seconds**. The code runs very quickly!